25 FISSION AND FUSION

Scientists have speculated about the nature of the atom for thousands of years, but it is only relatively recently that our current ideas were developed. In this chapter you will read about how our ideas about the structure of the atom have developed over the centuries and how we use nuclear energy to produce electricity.

LEARNING OBJECTIVES

- Know that nuclear reactions, including fission, fusion and radioactive decay, can be a source of energy
- Understand how a nucleus of U-235 can be split (the process of fission) by collision with a neutron, and that this process releases energy as kinetic energy of the fission products
- Know that the fission of U-235 produces two radioactive daughter nuclei and a small number of neutrons
- Describe how a chain reaction can be set up if the neutrons produced by one fission strike other U-235 nuclei
- Describe the role played by the control rods and moderator in the fission process
- Understand the role of shielding around a nuclear reactor
- Explain the difference between nuclear fusion and nuclear fission
- Describe nuclear fusion as the creation of larger nuclei resulting in a loss of mass from smaller nuclei, accompanied by a release of energy
- Know that fusion is the energy source for stars
- Explain why nuclear fusion does not happen at low temperatures and pressures, due to electrostatic repulsion of protons

NUCLEAR REACTIONS AS A SOURCE OF ENERGY

Nuclear reactions involve a change in the qualities of atoms. Heavy atoms may split into lighter atoms and other pieces in a process called fission. Lighter atoms may be forced to join together to make heavier atoms in a process called fusion. In either process, the mass of the starting atoms is greater than the mass of the products. This missing mass has been converted into energy.

Within the core of the Earth, radioactive isotopes of elements like uranium, thorium and potassium provide a large proportion of the heat within the Earth itself through radioactive decay.

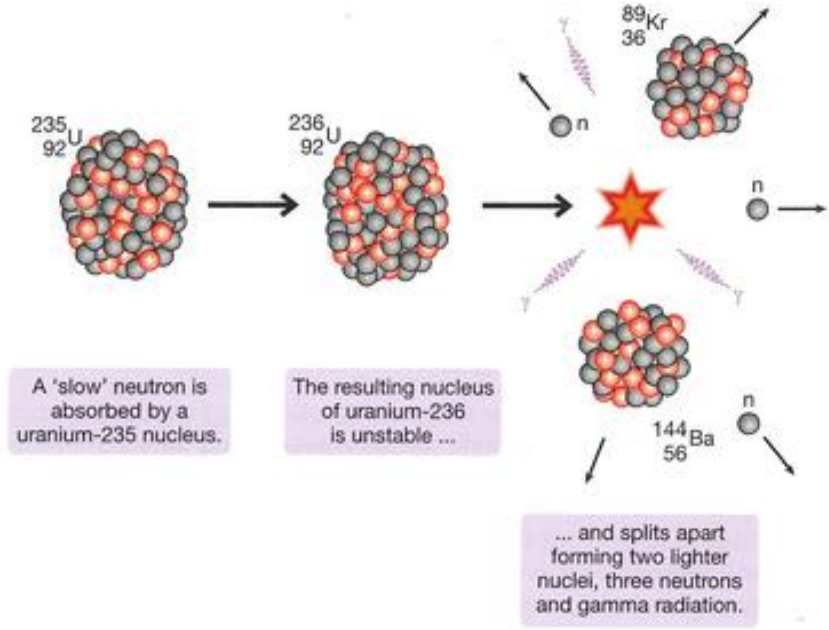
In the Sun, hydrogen is converted into helium in a **fusion reaction** providing us with a continuous supply of energy in the form of heat and other electromagnetic radiation.

NUCLEAR FISSION

Uranium-235 is used as fuel in a nuclear reactor. It is used because its nuclei can be split by a neutron. The process of splitting an atom is called fission.

Uranium-235 is called a fissile material because it goes through the splitting process easily. The fission process is shown in Figure 25.1.

In the fission reaction, a slow-moving neutron is absorbed by a nucleus of uranium-235.



▲ Figure 25.1 One example of fission of uranium-235

The resulting nucleus of uranium-236 is unstable and splits apart. The fragments of this decay are the two daughter nuclei of barium-144 and krypton-89. The decay also produces gamma radiation and three more neutrons. The equation for this decay is:

$$^{236}_{92}U \rightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3\,{}^{1}_{0}n + \gamma \text{ radiation}$$

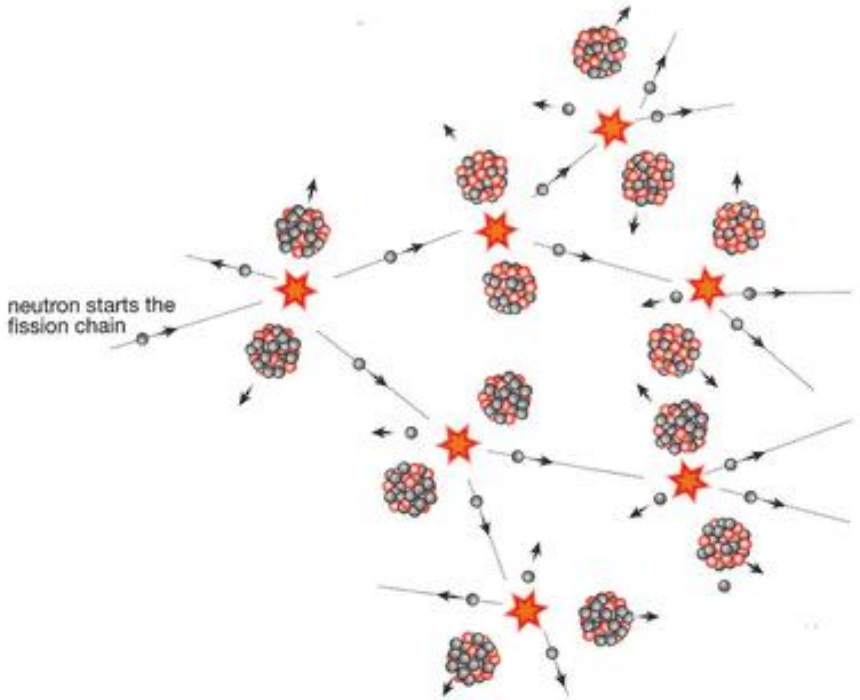
The fission reaction produces a huge amount of energy. This is because the mass of the products, the barium and krypton nuclei and the three neutrons, is slightly less than that of the original uranium-236 nuclei. This lost mass is converted to energy. Most of the energy is carried away as the kinetic energy of the two lighter nuclei. Some is emitted as gamma radiation. The three neutrons produced by the fission may hit other nuclei of uranium-235, so causing the process to repeat, as shown in Figure 25.2. If one neutron from each fission causes one nearby uranium-235 to split, then the fission reaction will keep going. If more than one neutron from each fission causes fission in surrounding nuclei, then the reaction gets faster and faster – a bit like an avalanche.

EXTENSION WORK

A 'slow' neutron is a low-energy neutron produced by a nuclear decay. Faster moving, more energetic neutrons do not cause fission.

This is called a chain reaction. Each fission results in more nuclei splitting apart. If the amount of uranium-235 is small, many of the neutrons released do not hit other uranium nuclei and the reaction does not get faster and faster. For a chain reaction to happen there must be a minimum amount of the uranium-235. This minimum amount is called the critical mass.

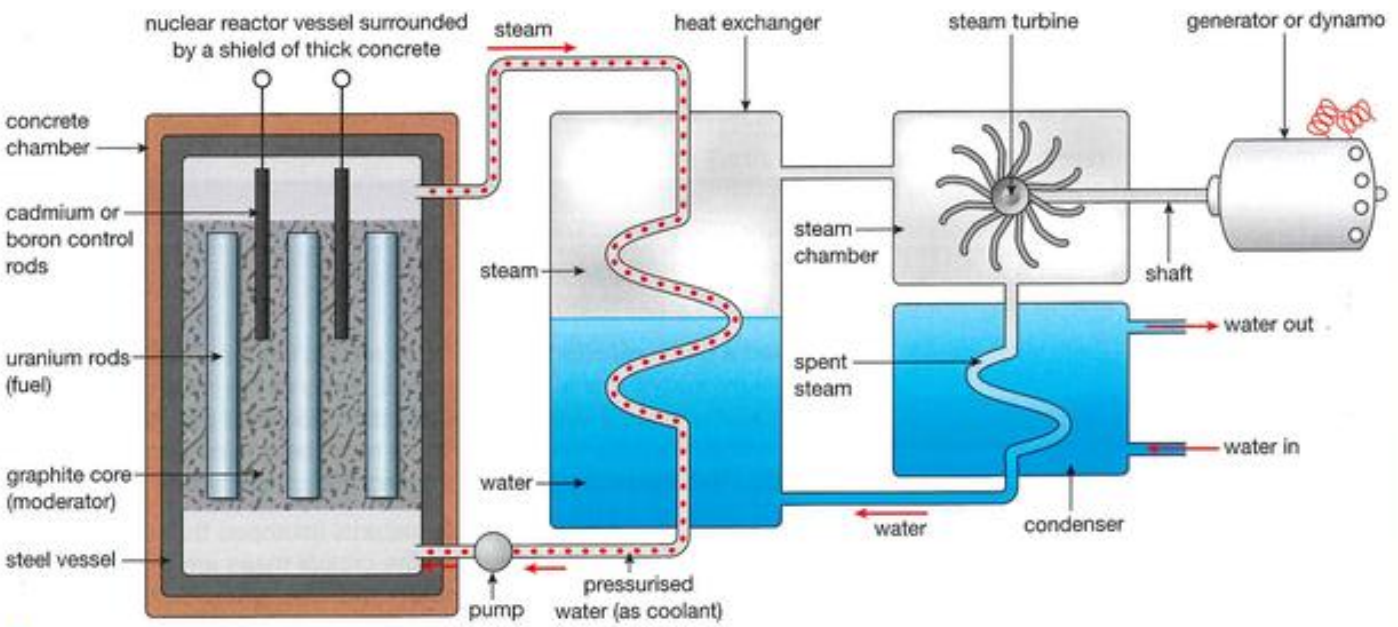
In an atomic bomb two pieces of fissile material (isotopes that can be triggered into splitting apart) that are smaller than the critical mass are forced together under high pressure to form a mass greater than the critical mass. The result is a chain reaction with the rapid and uncontrolled release of huge amounts of energy.



▲ Figure 25.2 A chain reaction in uranium-235

If this is allowed to take place in a nuclear reactor, the reactor core overheats, resulting in a nuclear explosion with the sudden release of enormous amounts of heat energy and radiation. In a nuclear reactor the process is controlled so that the heat energy is released over a longer period of time. The heat produced in the core or heart of the reactor is used to heat water. The steam produced then drives turbines (engines) to turn generators. The basic parts of a nuclear reactor are shown in Figure 25.3.

The reactor core contains fuel rods of enriched uranium. Enriched uranium is uranium-238 with a higher proportion of uranium-235 than is found in natural reserves of uranium. Graphite is used as a **moderator**. The job of



▲ Figure 25.3 A nuclear reactor controls a chain reaction so that heat energy is released slowly.

the moderator is to absorb some of the kinetic energy of the neutrons to slow them down. This is because slow neutrons are more easily absorbed by uranium-235. A neutron slowed in this way can start the fission process.

In the nuclear reactor there are also **control rods**, made of boron or cadmium. These absorb the neutrons and take them out of the fission process completely. When the control rods are fully inside the core, the chain reaction is almost completely stopped and the rate of production of heat is low. As the control rods are withdrawn, the rate of fission increases producing heat at a greater rate.

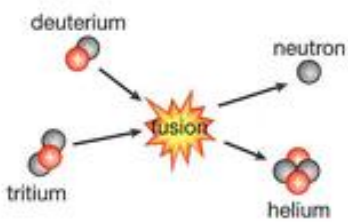
The reactor vessel is made of steel and surrounded by a concrete layer about 5 metres in thickness. This prevents any radiation escaping, even neutrons.

The nuclear process in a reactor produces a variety of different types of radioactive material. Some have relatively short half-lives and decay rapidly. These soon become safe to handle and do not present problems of long-term storage. Other materials have extremely long half-lives. These will continue to produce dangerous levels of ionising radiation for thousands of years. These waste products present a serious problem for long-term storage. They are usually sealed (closed) in containers which are then buried deep underground. The sites for underground storage have to be carefully selected. The rock must be water resistant and the geology of the site must be stable – storing waste in earthquake zones or areas of volcanic activity would not be sensible.

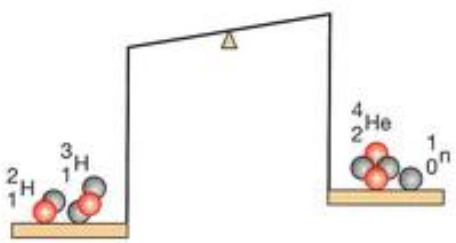
Some reactors are designed to produce plutonium. Plutonium is a very radioactive artificial element. Small amounts of plutonium represent a serious danger to health. Plutonium is another fissile material. If a large enough mass of plutonium is brought together a chain reaction will start. Plutonium can be used in the production of nuclear weapons.

Nuclear power stations do not produce carbon dioxide or acidic gases as fossil fuel power stations do. This means that nuclear power does not contribute to global warming or acid rain. Only small amounts of uranium are needed for a chain reaction and the supply of nuclear fuel will last many hundreds of years – unlike some fossil fuels that could run out in the next fifty years.

NUCLEAR FUSION



▲ Figure 25.4 Here a nucleus of deuterium collides with a nucleus of tritium. They undergo fusion to form the nucleus of helium, a neutron and a large amount of energy.



▲ Figure 25.5 The mass of the products of fusion is smaller than the two hydrogen nuclei.

Figure 25.4 shows a fusion process that is being used in projects to develop nuclear fusion reactors. Two isotopes of hydrogen, deuterium (H-2) and tritium (H-3), collide at very high speed. The result is the formation of a helium nucleus, a neutron and a large amount of energy. The fusion process is the energy source of our Sun and all stars.

This process uses materials that are more readily available than the uranium used in a conventional fission reactor and has the advantage of producing no radioactive waste. The problem is creating the very high temperatures needed to make the deuterium and tritium nuclei collide with enough energy to overcome the repulsive (pushing away) force between the positive electric charge in the nuclei of each isotope. The high temperature (100 million °C) liquid must be contained within a very strong magnetic field or 'magnetic bottle'. In order to increase the chance of fusion between the light nuclei in the hot liquid, the pressure within the liquid must be very high.

CHAPTER QUESTIONS

More questions on the structure of the atom can be found at the end of Unit 7 on page 255.

SKILLS CRITICAL THINKING (8)

1 a Uranium-235 (U-235) is a fissile material. What does this mean?

b If there is a large enough mass of U-235, it may cause a chain reaction. (This is called the critical mass for the isotope.)

(9) i Describe how a chain reaction may take place when a U-235 nucleus splits apart.

(10) ii A chain reaction is unlikely to take place if there is only a small amount of U-235. Explain why this is so.

(7) 2 List two advantages and two disadvantages of nuclear fission as a way of producing energy.

SKILLS REASONING

3 In a nuclear fission reactor, fuel rods made up of a fissile material, like U-235, are lowered into tubes in a large block of graphite. In between the fuel rods are control rods made of boron which may be raised out of the reactor or lowered into it. The reactor container (vessel) is surrounded by steel and a very thick layer of concrete.

(9) a Explain the purpose of the graphite around the fuel rods.

(8) b Describe what effect the action of raising the control rods out of the reactor core has on the behaviour of the reactor.

c Explain how the workers in a nuclear power station are protected from the dangerous types of radiation produced by this nuclear reactor.

SKILLS CRITICAL THINKING (9)

4 Describe the differences between nuclear fission and nuclear fusion.

5 Explain why creating energy in a nuclear reactor by fusion is so much more difficult than creating power by nuclear fission.